

3.18

a. $\widehat{\text{INSURANCE}} = 6.855 + 3.880 \text{ INCOME}$

slope = 3.880, Intercept = 6.855

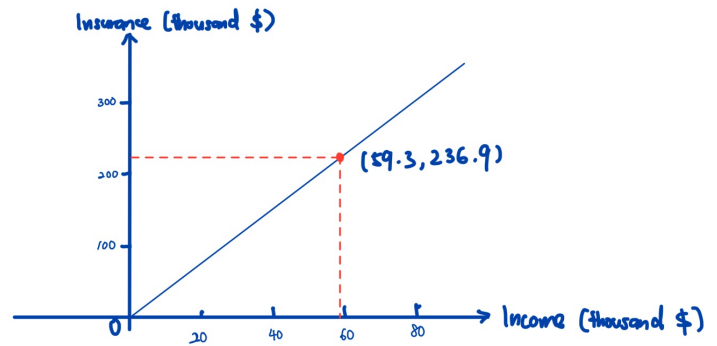
$se(b_1) = 7.383$, $se(b_2) = 0.112$

$\bar{x} = 59.3$

The OLS fitted line passes through $(\bar{x}, \bar{y})$.

$\hat{y} = 6.855 + 3.880(59.3) = 236.939$

The sample mean of insurance held is \$236939.



b. For each additional \$1000 in income, the average amount of insurance held is estimated to increased by \$3880.

$df = 20 - 2 = 18$, $\alpha = 0.05$, $t_{(0.025, 18)} = 2.101$

95% interval estimate,

$CI = b_2 \pm t_c se(b_2) = 3.880 \pm 2.101(0.112) = 3.880 \pm 0.2353$

$CI = [3.6447, 4.1153]$

In repeated sampling, about 95% intervals constructed this way will contain the true value of β_2 .

Therefore, we are 95% confident that for each additional \$1000 in income, the average amount of insurance held is estimated to increased by \$3880 between \$3645 and \$4115.

c. When $\text{INCOME}(x_0) = 100$

point estimate: $\widehat{\text{INSURANCE}} = 6.855 + 3.880(100) = 394.855$

$\widehat{\text{Cov}}(b_1, b_2) = -0.746$

$$\begin{aligned} \text{Var}(\hat{y}_0) &= \text{Var}(\hat{b}_1) + x_0^2 \text{Var}(\hat{b}_2) + 2x_0 \widehat{\text{Cov}}(\hat{b}_1, \hat{b}_2) \\ &= 7.383^2 + (100^2)(0.112)^2 + 2(100)(-0.746) \\ &= 30.7487 \end{aligned}$$

$se(\hat{y}_0) = \sqrt{30.7487} = 5.545$

99% $CI: t_{(0.005, 18)} = 2.878$

$\hat{y}_0 \pm t_c se(\hat{y}_0) = 394.855 \pm 2.878(5.545) = 394.855 \pm 15.9585$

$CI = [378.897, 410.813]$

In repeated sampling, about 99% intervals constructed this way will contain the true expected value.

Therefore, the expected amount of insurance held by a household with \$100000 income is between \$378897 and \$410813.

d. slope = 5, the claim corresponds to $\beta_2 = 5$

Hypothesis, $H_0: \beta_2 = 5$

$H_1: \beta_2 \neq 5$

$\alpha = 0.05$, $df = 18$

$$\text{test statistic: } t^* = \frac{b_2 - \beta_2}{\text{se}(b_2)} = \frac{3.880 - 5}{0.112} = -10$$

Rejection region: $|t| > 2.101$

Since $|-10| > 2.101$, so reject H_0 . The data does not support the claim, the actual increase is significantly less than \$5000.

e. Hypothesis, $H_0: \beta_2 = 1$

$H_1: \beta_2 > 1$

$\alpha = 0.01$, $df = 18$

$$\text{test statistic: } t = \frac{b_2 - \beta_2}{\text{se}(b_2)} = \frac{3.880 - 1}{0.112} \approx 25.71$$

Rejection region: $t > 2.552$

Since $25.71 > 2.552$, we reject H_0 .

The slope is significantly greater than 1 at the 1% level.

3.23. a. $\text{PRICE} = \alpha_1 + \alpha_2 \text{SQFT}^2 + e$

$$\text{Marginal effect: } \frac{dE(\text{PRICE})}{d\text{SQFT}} = 2\alpha_2 \text{SQFT}$$

$$\therefore \text{SQFT} = 20, \text{ marginal effect} = 2\alpha_2(20) = 40\alpha_2$$

$$\widehat{\text{PRICE}} = 93.5659 + 0.1845 \text{SQFT}^2$$

(se) (6.0723) (0.005256)

① Hypothesis, $H_0: 40\alpha_2 \leq 13$

$H_1: 40\alpha_2 > 13$

$$\text{② Test statistic: } t^* = \frac{40\hat{\alpha}_2 - 13}{\text{se}(40\hat{\alpha}_2)} = \frac{40\hat{\alpha}_2 - 13}{40\text{se}(\hat{\alpha}_2)} = \frac{40(0.1845) - 13}{40(0.005256)} = -26.7288$$

③ $\alpha = 0.05$ (right-tailed), $df = 500 - 2 = 498$

④ Rejection region: $t_{(0.05, 498)} > 1.648$

⑤ $p\text{-value} = 1$

Because $p > \alpha$, and t^* falls in the non-rejection region, we do not reject H_0 .

There is insufficient evidence at the 5% significance level to conclude that the marginal effect of increasing a 2000 square foot house by 100 square feet is greater than \$13000.

```
Call:
lm(formula = price ~ sqft2, data = collegetown)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-383.66930 -48.39258  -7.50040   38.75211  469.70077
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 93.565854100  6.072225591  15.40882 < 2.22e-16 ***
sqft2       0.184519003  0.005255801  35.10769 < 2.22e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 92.07946 on 498 degrees of freedom
Multiple R-squared:  0.7122301, Adjusted R-squared:  0.7116523
F-statistic: 1232.55 on 1 and 498 DF, p-value: < 2.2204e-16
```

```
> cat("t-statistic:", t_a, "\n")
t-statistic: -26.72875166
> cat("rejection region: t >", t95_right, "\n")
rejection region: t > 1.647919139
> cat(" p-value: ", p_val_a)
p-value: 1
```

b. Marginal effect = $2\alpha_2(40) = 80\alpha_2$

① Hypothesis, $H_0: 80\alpha_2 \leq 13$

$H_1: 80\alpha_2 > 13$

② test-statistic: $t^* = \frac{80\hat{\alpha}_2 - 13}{\text{se}(80\hat{\alpha}_2)} = \frac{80(0.1895) - 13}{80(0.005256)} = 4.1895$

③ $\alpha = 0.05$ (right-tailed)

④ Rejection region: $t > 1.6479$

⑤ p-value $\approx 1.6539 \times 10^{-5}$

Because $p < 0.05$, t^* falls in the rejection region, we reject H_0 .

There is sufficient evidence at the 5% significance level to conclude that the marginal effect of increasing a 4000 square foot house by 100 square feet is greater than \$13000.

```
> cat("t-statistic:", t_b, "\n")
t-statistic: 4.189466715
> cat("rejection region: t >", t95_right, "\n")
rejection region: t > 1.647919139
> cat(" p-value: ", p_val_b)
p-value: 1.653949801e-05>
```

c. 2000 sqft $\Rightarrow$ SQFT = 20

$E(\widehat{PRICE|SQFT}) = \hat{\alpha}_1 + \hat{\alpha}_2 \text{SQFT}^2 = 93.5659 + 0.1845(20^2) = 167.375$

95% interval estimate: $\widehat{E}(\widehat{PRICE|SQFT}) \pm t_{(0.025, 498)} \text{SE}[\widehat{E}(\widehat{PRICE|SQFT})]$
 $= 167.3735 \pm 1.9647(4.7464)$

$CI = [158.0481, 176.6988]$

In repeated sampling, about 95% intervals constructed this way will contain the true expected value.

Thus, the expected price of a 2000 sqft house is between about \$158048 and \$176699.

```
      fit      lwr      upr
1 167.3734553 158.048061 176.6988495
```

d. In the sample, there are 3 houses with 2000 square feet of living area.

They sold for \$138000, \$169000 and \$183000. The average of the selling price is \$163333.

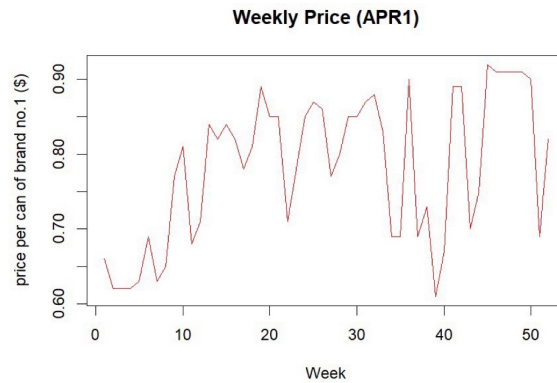
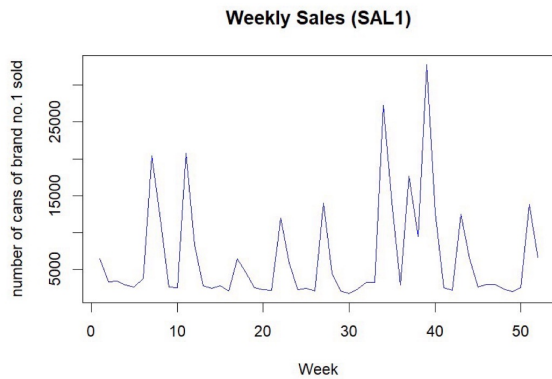
The results are highly compatible because the sample average of \$16333 falls within the 95% confidence interval [\$158048, \$176699].

```
> cat("number of houses: ", n20, "\n")
number of houses: 3
> cat("selling prices:", subset_20$price, "\n")
selling prices: 138 169 183
> cat("Sample Mean:", mean_price_20)
Sample Mean: 163.3333333>
> ## Check compatibility with (c)
> mean_price_20 >= ci_c[2] && mean_price_20 <= ci_c[3]
[1] TRUE
```

3.31

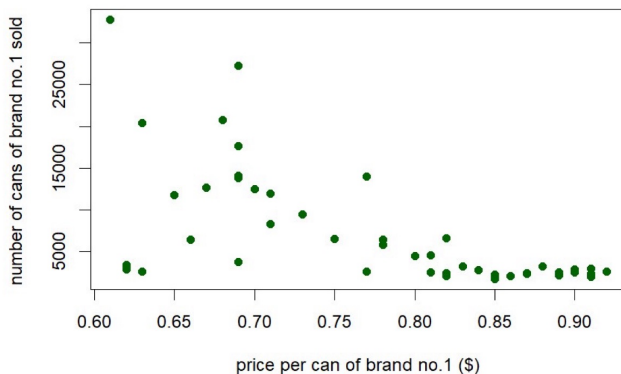
a. Summary Statistic

	Mean	Min	Max	Std. dev
SAL1	6718.7115	1762	32820	6915.083
APR1	0.7825	0.61	0.92	0.0980



Based on the plots, there is substantial variation in both sales and price from week to week. There are sharp, periodic drops in the price of tuna. Corresponding with these price drops, there are massive spikes in sales volume.

Sales vs Price(Weekly)



There is an inverse relationship between weekly sales and weekly price. Yes, this is what we would expect from demand theory because a higher price is associated with lower quantity demanded.

C. $\widehat{SAL} = 40714.22 - 434.45 \text{ PRICE1}$

We estimate that a one cent increase in the price of brand no. 1 will reduce the expected weekly sales of brand no. 1 by 434.45 cans

The 95% interval estimate for this effect is $[-592.2897, -276.6049]$. In repeated sampling, about 95% interval constructed this way will contain the true value of β_1 . Thus, a one-cent increase in price is estimated to reduce weekly sales by about 276.60 to 592.29 cans.

```
> cat(sprintf("Point Estimate for beta2: %.4f\n", b2))
Point Estimate for beta2: -434.4473
> cat(sprintf("95% Interval Estimate for beta2: [%.4f, %.4f]\n", ci_c_lower, ci_c_upper))
95% Interval Estimate for beta2: [-592.2897, -276.6049]
```

```
Call:
lm(formula = sal1 ~ price1, data = tuna)

Residuals:
    Min       1Q   Median       3Q      Max
-10869.5  -1588.3   -499.3   1626.2  18607.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40714.22   6196.42   6.571 2.82e-08 ***
price1       -434.45    78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,    Adjusted R-squared:  0.367
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

d. $E(\widehat{SAL} | \text{PRICE1} = 70) = 10302.9$

90% interval estimate for expected sales: $[8624.93, 11980.87]$

Thus, when the price per can is 70 cents, the 90% interval estimate for the expected number of cans sold in a week is between 8624.93 and 11980.87.

```
> newd <- data.frame(price1 = 70)
> ci_d <- predict(mod, newdata=newd, interval="confidence", level=0.90)
> print(ci_d)
      fit      lwr      upr
1 10302.9 8624.934 11980.87
```

e. $\hat{\epsilon} : \hat{\beta}_1 \cdot \frac{\bar{x}}{\bar{y}} = -434.4473 \times \frac{78.25}{6718.71} = -5.0598$

95% interval estimate for elasticity: $[-6.8981, -3.2215]$

At the sample means, a 1% increase in price is associated with about a 5.06% decrease in sales. Since the absolute value is greater than 1, the sales are fairly elastic with respect to price. This can make economic sense if canned tuna has close substitutes and consumers respond strongly to price changes.

```
> cat(sprintf("Point Estimate of Elasticity: %.4f\n", elasticity))
Point Estimate of Elasticity: -5.0598
> cat(sprintf("95% Interval for Elasticity: [%.4f, %.4f]\n", ci_e_lower, ci_e_upper))
95% Interval for Elasticity: [-6.8981, -3.2215]
```

f. Hypothesis: $H_0: \varepsilon = -3$, $H_1: \varepsilon \neq -3$

$$\Rightarrow H_0: \beta_2 = -3 \times \frac{\overline{SAL1}}{\overline{PRICE1}} = -3 \times \frac{6718.71}{78.25} = -257.59$$

$$H_1: \beta_2 \neq -257.59$$

$$\alpha = 0.10 \text{ (two-tailed)}, df = 52 - 2 = 50$$

$$t\text{-statistic} = \frac{-434.4473 + 257.59}{78.584} = -2.2506$$

$$\text{Rejection region: } |t| > 1.6759$$

$$p\text{-value} \approx 0.0288$$

Because $|t| = 2.2506 > 1.6759$ and $p\text{-value} < 0.10$, we reject H_0 .

There is sufficient evidence at the 10% significance level to conclude that the elasticity is not minus three.

```
> cat(sprintf("Test Statistic (t): %.4f\n", t_stat))
Test Statistic (t): -2.2506
> cat(sprintf("Rejection Region: |t| > %.4f\n", t90_two))
Rejection Region: |t| > 1.6759
> cat(sprintf("P-value: %.4e\n", p_val))
P-value: 2.8842e-02
>
> if(p_val < 0.10) {
+   cat("Reject H0.\n")
+ } else {
+   cat("Do not reject H0.\n")
+ }
Reject H0.
```
